

GAME DESIGN DOCUMENT

PaliPali

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Applied Tools & Project 1 Video

Summary of the tools used to build PaliPali:

Game engine / source code: Unity 6.5 (6000.5), scripted in C#, edited in Visual Studio Code with the C# Dev Kit extension.

Graphics: base background and character-run asset pack from YouTuber Nomad Coder <https://on.unitysquare.co.kr/3ZMF3O7>. The playable character artwork was generated with Google Gemini and had its background removed with <https://remove.bg> before import into Unity.

SFX / Music: background and intro music are two tracks from DanAndPhil's copyright-free "All Beats" collection - "You Won't Believe What Happened" (background loop) and "Ready Player Two" (plays when a run starts). Five sound effects (jump, hit, eat, power-up, game over) were custom-made with <https://sfxr.me/>

Version control: Git, hosted on GitHub, managed with GitHub Desktop (repository: <https://github.com/hyechanshin/Runner-Man-2D-Unity-Game-Project>).

Gameplay video: <https://youtu.be/MnwQKv1vyFM>

Playable build: <https://play.unity.com/en/games/a0448229-50a7-46d4-8f90-7574e6229f9e/palipali>

1. Executive Summary, Quick overview

PaliPali is a 2D side-scrolling endless runner built in Unity, set in a Korean-flavored cityscape. The player runs automatically past a scrolling skyline and jumps to dodge oncoming enemies, a delivery rider and a fast-moving grandmother ("Halmoni"), while collecting Korean food items (gochu chili peppers and baechu cabbages) that restore health, plus a rare golden baechu that grants temporary invincibility. The player starts with a limited number of hearts, loses one on enemy contact, and the run ends when hearts reach zero. Survival time, scaled by a chosen difficulty, becomes the score, with food pickups adding bonus points and a persistent high score saved per difficulty. The name comes from the Korean phrase 빨리빨리 (ppalli-ppalli, "hurry hurry"), matching the game's ever-accelerating pace.

2. Target Audience

PaliPali is aimed at casual players who enjoy short, pick-up-and-play arcade experiences similar to classic endless runners. Adjustable EASY / NORMAL / HARD difficulty and a choice of two playable characters make it approachable for newcomers while still rewarding skilled play, and built-in tap-input support means it's playable on both desktop and touch devices.

3. Main Characters

The player chooses between two playable runners on the title screen (Up/Down arrows): RUNNER, a male character with 3 hearts and a jump force of 15, and RUNNER F, a female character with 2 hearts but a higher jump force of 20 - trading survivability for more precise obstacle clearance. Neither character has combat abilities; their only actions are running (automatic) and jumping.

4. Main Features

4.1 Main mechanics

The core loop: keep running, jump to dodge enemies, and collect food to stay alive. The player has a set number of hearts (3 or 2, depending on the chosen character); colliding with an enemy costs one heart, colliding with a food item (gochu or baechu) restores one heart, and colliding with the rare golden baechu grants temporary invincibility. The run ends when hearts reach zero. Score is based on survival time scaled by the chosen difficulty's multiplier, plus one bonus point per food item collected. Difficulty (EASY / NORMAL / HARD, chosen with Left/Right arrows) sets the starting speed, maximum speed, ramp rate and score multiplier, and each difficulty tracks its own separate high score via PlayerPrefs.

4.2 Movement

The character moves automatically along a single horizontal axis; the player's only input is jumping (Space bar, mouse click, or touch tap, unified through a shared TapInput script). Pressing jump while grounded launches the character upward with a jump force set by the chosen character. Obstacles, enemies and food scroll toward the player at a speed that ramps up smoothly over time ($\text{speed} = \text{base} + \text{elapsed} \times \text{rampRate}$, clamped to a difficulty-specific maximum), so the game gets harder the longer a run lasts.

4.3 Physics

Movement and collision are handled through Unity's built-in 2D physics engine. The player uses a Rigidbody2D and jumps via an impulse force (AddForce with ForceMode2D.Impulse) combined with Unity's default gravity for a natural arc; a ground-check flag (isGrounded), set via OnCollisionEnter2D against the platform, prevents jumping again mid-air. Enemies and food items use trigger colliders (OnTriggerEnter2D, Is Trigger enabled) tagged by type, enemy, food, golden - so the player script's Hit(), Heal() and StartInvincible() functions can react to each one differently.

4.4 Multiplayer Mode

None - PaliPali is a single-player experience only.

5. Genre, Setting, Concept Art book*

PaliPali is a 2D side-scrolling endless runner set in a stylized, Korean-flavored pixel-art cityscape. Three parallax background layers - sky, buildings, and a street-level platform - scroll at different speeds to create a sense of depth and constant forward motion



6. Enemies, NPCs, Other objects

PaliPali features two enemy types that spawn and scroll toward the player: a delivery-rider enemy and a fast-moving Halmoni (grandmother) enemy, both of which cost the player a heart on contact. Three collectible food objects counter these: gochu (chili pepper) and baechu (cabbage), which each restore one heart, and a rare golden baechu, which grants the player temporary invincibility. Buildings/obstacles, enemies and food are each driven by a shared Spawner / Mover / Destroyer script trio living on dedicated spawner objects (a building spawner, an enemy spawner, and food/gold spawners), with every prefab tagged by type so the player's trigger-collision logic can respond correctly.

7. Story board, script*

7.1 Story overview

PaliPali doesn't carry a deep narrative, in keeping with its small arcade scope: the player is simply trying to get through a hectic, obstacle-filled city as fast, and for as long as possible, in the spirit of 빨리빨리 ("hurry, hurry"), while dodging a delivery rider and a determined grandmother, and grabbing food along the way to stay standing.

7.2 Progression, World 1

Rather than discrete worlds or stages, PaliPali is a single continuous run whose challenge escalates smoothly over time: speed ramps up the longer the player survives ($\text{base} + \text{elapsed} \times \text{rampRate}$, capped at a difficulty-specific maximum). The player sets their starting challenge level up front via the EASY / NORMAL / HARD difficulty selector on the title screen, each with its own starting speed, ramp rate, max speed and score multiplier.

8. Technical definitions, Tech guide*

8.1 Platforms, versions

Built in Unity 6.5 (6000.5) using C#, and deployed as a browser build via Unity Play - playable directly at <https://play.unity.com/en/games/a0448229-50a7-46d4-8f90-7574e6229f9e/palipali> in a standard browse, with no installation or developer mode required. Touch input is supported in-browser, though no dedicated Android build has been produced.

8.2 Control Scheme

Keyboard: Space bar to jump / start / restart; Left/Right arrows to pick a difficulty; Up/Down arrows to pick a character on the title screen. Mouse click and touch tap are unified with the space bar through a shared TapInput script, so jumping, starting, restarting, and difficulty/character selection are all also fully usable on touch devices

8.3 Limitations

As a course project, PaliPali is scoped to a single continuous endless-runner level rather than multiple discrete levels or worlds, and does not include a dedicated Android build even though the interface is touch-compatible in-browser. Progress is not saved beyond the high score, which persists locally per difficulty via PlayerPrefs.

9. Business Definitions

9.1 In-app Purchases

None. PaliPali is a free, non-commercial student project.

9.2 DLC Packs

None planned.

10. Outsourced/Bought Assets

Every asset used that was not made entirely from scratch, with source and license notes:

- Background and base character/animation asset pack - YouTuber Nomad Coder (<https://on.unitysquare.co.kr/3ZMF3O7>).
- Player character artwork - base image generated with Google Gemini (AI image generation), background removed with <https://remove.bg>.
- Background music - "You Won't Believe What Happened" and "Ready Player Two" by DanAndPhil, from their copyright-free "All Beats" collection (<https://youtu.be/u6xy-vXlGIY>, <https://youtu.be/OTeHBM7h7Hw>).
- Sound effects - 5 custom SFX (jump, hit, eat, power-up, game over) created with the <https://sfxr.me> tool.
- Unity Input System and TextMeshPro - built-in Unity packages (Unity Technologies), used for jump/tap input handling and the score UI respectively.